Form AR 140

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5207
Subject Title	Compressible Aerodynamics
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To equip students with a profound understanding of compressible aerodynamics, emphasising the behaviour of gases at high speeds. 2. To develop students' capability to analyse canonical geometries with compressibility effects.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. acquire fundamental knowledge of compressible aerodynamics; b. perform theoretical analysis of canonical flow problems; and c. be exposed to state-of-the-art research advances.
Subject Synopsis/ Indicative Syllabus	**One-dimensional and quasi-one-dimensional flows** – Normal shock relations; Area-velocity relation; Nozzles. **Oblique shock and expansion waves** – Oblique shock relations; Conical flow; Prandtl-Meyer expansion waves. **Linearized flow** – Velocity potential equation; Linearised subsonic flow; Compressibility corrections; Linearised supersonic flow. **Transonic and hypersonic flows** –Full velocity potential equation; Newtonian theory; Mach Number independence. **Boundary-layer flows** – Boundary-layer theory; Self-similar solutions. **Case studies** – Theoretical analysis of canonical flow problems in aerospace engineering.
Teaching/Learning Methodology	1. The teaching and learning methods include lectures/tutorials, projects, and homework assignments. 2. The lectures/tutorials aim at providing students with integrated knowledge of compressible aerodynamics. 3. Technical/scientific problems and examples will be raised in projects and homework assignments to develop students' skills of aerodynamic analysis.

Teaching/Learning Methodology	Intended subject learning outcomes		
	a	b	c
1. Lectures/tutorials	√	√	√
2. Projects	√	√	√
3. Homework assignments	√	√	√

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
		a	b	c
1. Projects	30%	√	√	√
2. Tests	20%	√	√	
3. Final exam	50%	√	√	
Total	100%			

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

1. The assessment is comprised of 50% continuous assessment (projects and tests) and 50% examination.
2. The continuous assessment consists of projects and tests. They are used to evaluate the progress of students' study, assist them in self-monitoring of fulfilling the respective subject learning outcomes, and enhance the integration of the knowledge learnt.
3. The examination is used to assess the knowledge acquired by the students for understanding and analysing the problems critically and independently; as well as to determine the degree of achieving the subject learning outcomes.

Student Study Effort Expected

Class contact:	
▪ Lecture	33 Hrs.
▪ Tutorial	6 Hrs.
Other student study effort:	
▪ Self-learning	33 Hrs.
▪ Homework	50 Hrs.
Total student study effort	122 Hrs.

Reading List and References

1. Anderson J. D., Fundamentals of Aerodynamics. McGraw-Hill, 6th edition, 2016. ISBN 13: 978-1259129919
2. Anderson J. D., Modern Compressible Flow: With Historical Perspective.

	McGraw-Hill, 3rd edition, 2012. ISBN 13: 978-0072424430 3. Bertin J. J. and Cummings R. M., Aerodynamics for Engineers. Pearson, 6th edition, 2013. ISBN 13: 978-0132832885

Jun 2024